

What Does the Image Add?

Text-Conditioned Visual Detail Completion

CSCI-GA 2271 Computer Vision Fall 2026 Revised Proposal

Track: Agent Track Format: Research Project

September 29, 2026

1 Research Question and Significance

A short caption often identifies an object without describing details a reader may need. Given an image, existing text, and a specified instance, can a model add a concise, correct detail that the text does not already convey? For example, after “a person is cycling,” a useful supplement might be “wearing a red helmet.” If the helmet is already mentioned, repeating it adds little; assigning another person’s helmet to the cyclist is an error.

We study **instance-grounded visual detail completion**. The immediate use case is enriching an existing description for a reader interested in a particular object. We will measure whether the supplement helps answer independently written image questions. This is a testable usefulness hypothesis, not evidence of an already validated application.

Our questions are: **RQ1**, can grounded supervision improve descriptions of the correct instance? **RQ2**, does conditioning generation on existing text improve genuinely missing detail recovery over ordinary region and full-image captioning? **RQ3**, do any gains survive factual auditing, automatic localization, and evaluation on a different dataset?

2 Completed Evidence

Earlier experiments generated full-image and regional descriptions and then selected a supplement. Positive source-domain proxy results did not transfer reliably, and learning to choose among candidate sentences often failed. We therefore moved supervision into *grounded generation*, using human phrase-to-box links rather than asking a model to certify its own output.

The completed E100 experiment fine-tuned Qwen2.5-VL-3B for one epoch with LoRA on 85,609 Flickr30k Entities phrases. It receives a marked image, but *no existing text*. On 20,544 held phrases from 5,300 image-disjoint images, it improved reference-phrase recovery under two primary automated evaluators and a prespecified third-reader sensitivity analysis:

Evaluator	Frozen	Trained	Gain (pp)
DeBERTa NLI	22.10	38.41	+16.32
Smol text reader	45.43	64.67	+19.24
Qwen7 sensitivity	30.53	42.64	+12.12

All gains have positive image-cluster bootstrap intervals. Masking the marked region with gray substantially reduced performance. These are **automated phrase-recovery scores, not open-caption factual accuracy**.

The fixed adapter’s Visual Genome transfer test failed its preregistered two-evaluator gate: Smol’s gain was only +2.00 points, with a 95% interval of $[-2.50, +6.25]$, despite a pos-

itive DeBERTa result. An initial existing-text evaluation is positive under DeBERTa but awaits its second reader. Thus we have evidence for a grounded-description component, not yet for text-conditioned completion, reliable transfer, or end-to-end utility. Results are frozen to the [September 29 repository snapshot](#).

3 Related Work and Contribution

Flickr30k Entities provides cross-caption entity links and phrase grounding [1]; Visual Genome provides regional descriptions and QA annotations [2]. FineCaption already studies controlled regional descriptions [3]. CapWAP learns captions for information needs using QA rewards [4], while PromptCap conditions captioning on a question [5]. Consequently, region captioning, LoRA, or a caption-to-QA improvement alone is not a new contribution.

Our course contribution will be a reproducible study of **what grounded supervision fixes and what remains unsolved when completing existing text**. The publication hypothesis is narrower: learning a supplement that changes with the reader’s existing knowledge while preserving instance correctness, without seeing the downstream question. It requires mechanism ablations and useful gains over strong alternatives; novelty is not assumed.

4 Proposed Method

Let I be the image, T its existing description, e the entity mention in T , and b the target box. A vision-language generator predicts a short supplement $s = g_\theta(I, b, T, e)$. Initially b is supplied by an annotation or a user’s selection. Automatic localization is a separate extension, so failures in finding the object can be distinguished from failures in describing it.

Annotation-based training. Across captions of the same Flickr30k image, we identify a phrase describing the same entity as e but containing a content token absent from T . This deterministic rule provides candidate residual targets, not guaranteed semantic novelty. The prepared E107 set contains 63,786 train targets over 19,626 images and 15,300 held targets over 4,720 different images. We will train one fixed LoRA configuration with target-token cross-entropy, then score the frozen held set without a checkpoint sweep.

Mechanism checks. E100 and E107 use different training sets, so a subsequent matched-data, matched-budget ablation will remove T while keeping targets fixed. Adding a verified fact to T should reduce repetition of that fact; paraphrasing T should preserve useful content. Same-image wrong-instance and gray-region controls test whether the correct visual evidence matters.

Automatic localization. A frozen binding model will score a finite hierarchy of windows using T and e , without the target phrase or annotated box. Report localization quality and retention of human-box description gains. This is a component evaluation, not a proposed new detector.

5 Evaluation

Data and splits. Flickr30k Entities is the main development setting; Visual Genome tests transfer. Split by image, keep all phrases and text states of an image together, and check exact and perceptual duplicates. Previously inspected examples remain development data. Official test and reserved confirmation identities stay sealed until the method, metrics, sample size, and comparison plan are frozen.

Baselines. Compare text alone; frozen and trained marked-region descriptions; full-image generic and entity-directed descriptions; and the frozen prompt-only version of our text-conditioned model. To distinguish a method gain from fine-tuning alone, add a full-image control trained on the same examples and budget. Fix supplement length for the main comparison, and also report stronger full-image outputs allowed a larger budget. Purposeful-caption and question-conditioned PromptCap-style controls will contextualize utility; the latter have an explicitly disclosed question-access advantage. Adaptations will be identified as such rather than called official reproductions.

Metrics. Report (i) reference-phrase recovery and missing-detail recall; (ii) visual factual precision, wrong-instance and contradiction rates; (iii) repetition of T , abstention, and text-state sensitivity; (iv) text-only QA accuracy from $T + s$, including answers harmed by the supplement; and (v) total generation/evaluation time, tokens, memory and allocated GPU-hours. Selected region area is not compute cost. Cluster uncertainty estimates by image and report every prespecified reader and seed.

Verification. Two blinded model families review facts and QA quality, with positive, wrong-box and answer-leak controls. Failed reviewers cannot establish a pass. Preserve uncertainty and label these results automated proxies. Use multiple valid references where available; a different correct description need not match one reference. Plan an image-visible human audit before claiming human-valid quality.

Practical-value gate. Independently written Visual Genome questions must be connected to objects through official QA-to-region mappings, not shared answer words. The current 512-image development packet is awaiting a blinded quality audit; it is not a completed utility experiment. Only eligible data and a successful residual generator proceed to a frozen text-only answerer. We measure information conveyed by the supplement while hiding questions from generation. Answer gain must exceed introduced errors at an acceptable cost.

Decision rule. The registered E107 development gate requires at least 3 percentage points in reference recovery from T plus the output over each of its three fixed baselines, with positive paired intervals under both primary readers. It licenses further evaluation, not a publication claim. Broader utility and confirmation endpoints will be fixed prospectively. A failure

triggers error analysis and a materially new hypothesis; we will not retune against the same holdout until it passes.

6 Deliverables, Risks, and Timeline

Deliverables. Reproducible data, training and evaluation pipelines; grounded/text-conditioned comparisons and failure analyses; a 4–9-page LaTeX research report excluding references, plus an Agent Log appendix; and a code repository with incremental history. This proposal copy does not replace that history. An image/text/box viewer is a stretch goal; finance is outside scope.

Risks and responses. Audit semantic novelty rather than word absence. Check reviewers for bias and label collapse with independent controls. Report human and automatic boxes separately and retain negative transfer results. If strong full-image baselines match our method at lower cost, the course outcome will be a rigorous comparative study, not an unsupported superiority claim.

Preliminary schedule (relative to approval of this revision; align calendar dates with the course deadlines):

- **Weeks 1–2:** Complete the fixed seed replication, existing-text evaluation, strong full-image controls, and blind factual audit; freeze the error taxonomy.
- **Weeks 3–4:** Run the prepared text-conditioned generator and matched ablations; evaluate automatic localization only under its registered protocol.
- **Weeks 5–6:** Audit independent QA data; if gates pass, evaluate utility and transfer. Otherwise diagnose and register one material redesign.
- **Final two weeks:** Freeze claims and figures, finalize the report and Agent Log, and rehearse team code walkthroughs. Publication expansion is optional.

Resources and agent workflow. Use one available NYU HPC GPU within the reconciled budget, with caching, checkpoints, five-second monitoring and a prepared task queue. Agents assist with code, literature and analysis. Students verify decisions and results and prepare to explain all submitted code. The final report will attribute individual contributions; its Agent Log will document tools, division of work, agent failures, human diagnosis and verification.

References

- [1] B. A. Plummer et al. Flickr30k Entities: Collecting Region-to-Phrase Correspondences for Richer Image-to-Sentence Models. *IJCV*, 2017. [arXiv:1505.04870](#).
- [2] R. Krishna et al. Visual Genome: Connecting Language and Vision Using Crowdsourced Dense Image Annotations. *IJCV*, 2017. [arXiv:1602.07332](#).
- [3] H. Hua et al. FINECAPTION: Compositional Image Captioning Focusing on Wherever You Want at Any Granularity. *CVPR*, 2025. [arXiv:2411.15411](#).
- [4] A. Fisch et al. CapWAP: Image Captioning with a Purpose. *EMNLP*, 2020. [ACL Anthology](#).
- [5] Y. Hu et al. PromptCap: Prompt-Guided Task-Aware Image Captioning. *ICCV*, 2023. [arXiv:2211.09699](#).